

Two Prime Closed Geodesics on a Katok–Zermelo Randers Sphere

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Abstract

We give a cutoff-free closed-geodesic classification for rotational Zermelo navigation on the unit round two-sphere. An irrational wind leaves exactly the two oriented equators, with periods $2\pi/(1 \pm \epsilon)$ and explicit transverse Poincaré determinants. The rational-resonance, zero-wind, wind-reversal, degeneracy, and strong-convexity faces are closed without identifying generic and equatorial least periods.

1 Navigation convention

Let h be the round metric of curvature one on S^2 , with longitude of period 2π . Let $W = \epsilon \partial_\phi$ and $0 < |\epsilon| < 1$. Since $|W|_h \leq |\epsilon| < 1$, translating the h -unit circle in each tangent plane by W defines a strongly convex Randers metric F_ϵ . We orient its geodesics and use the clock $F_\epsilon(\dot{\gamma}) = 1$. The signs $+$ and $-$ refer to the two directions of the round equator before adding the wind.

On the nonzero cotangent bundle T^*S^2 , the dual navigation Hamiltonian is

$$F_\epsilon^*(x, \xi) = |\xi|_h + \xi(W). \quad (1)$$

The convention in (1) is equivalent to the translated indicatrix convention above and fixes every sign in the period formulas.

Lemma 1 (commuting flow factorization). *Let Φ_t be rotation generated by W . Every unit F_ϵ -geodesic, and no other curve, has the form*

$$\gamma(t) = \Phi_t(\rho(t)), \quad (2)$$

where ρ is a unit-speed round great circle. Moreover, F_ϵ has constant flag curvature one.

Proof. The cotangent lift of W preserves the round cometric because W is Killing. Hence the two homogeneous Hamiltonians in (1) Poisson commute. Their flows compose in either order: the first projects to round unit cogeodesic motion and the second to Φ_t . This proves (2) and its converse. The navigation curvature theorem for a Killing wind on a curvature-one background gives flag curvature one. $\square$

2 The complete irrational ledger

Theorem 2 (two-geodesic theorem). *If $\epsilon \notin \mathbb{Q}$, the complete oriented prime closed-geodesic ledger of (S^2, F_ϵ) consists of exactly the two equators. Their least periods are*

$$T_+ = \frac{2\pi}{1 + \epsilon}, \quad T_- = \frac{2\pi}{1 - \epsilon}. \quad (3)$$

In canonical transverse Jacobi coordinates their Poincaré maps have multipliers $e^{\pm iT_+}$ and $e^{\pm iT_-}$, respectively, and

$$\det(I - P(T)) = 2 - 2 \cos T = 4 \sin^2(T/2). \quad (4)$$

Both prime orbits are nondegenerate.

Proof. The oriented round unit tangent bundle is $SO(3)$; represent a frame by g . If $R_z(\alpha)$ is rotation through α about e_3 , round geodesic motion is $gR_z(t)$ and the wind acts by $R_z(\epsilon t)g$. Consequently a return at $T > 0$ is exactly

$$R_z(\epsilon T)gR_z(T) = g, \quad R_z(\epsilon T) = gR_z(-T)g^{-1}. \quad (5)$$

The rotation on the left of the second equality has axis e_3 ; the conjugated right rotation has axis ge_3 . At a non-equatorial frame these unoriented axes are distinct. Thus equality forces both rotations to be the identity, including at angle π , where a nontrivial rotation still determines its unoriented axis. Hence

$$T = 2\pi m, \quad \epsilon T = 2\pi n, \quad m > 0, \quad m, n \in \mathbb{Z}. \quad (6)$$

This is impossible for irrational ϵ .

The exceptional frames are precisely the two equators, for which $ge_3 = \pm e_3$ and the unoriented axes coincide. Their signed angular velocities are $1 + \epsilon$ and $-1 + \epsilon$. Since $|\epsilon| < 1$, their positive least periods are exactly (3).

Flag curvature one makes a scalar transverse Jacobi coefficient satisfy $j'' + j = 0$. In the returning canonical normal frame its fundamental matrix is

$$P(T) = \begin{pmatrix} \cos T & \sin T \\ -\sin T & \cos T \end{pmatrix}.$$

This gives the characteristic polynomial $z^2 - 2(\cos T)z + 1$ and (4). If an irrational-wind equator were degenerate, $T_{\pm}/(2\pi) = 1/(1 \pm \epsilon)$ would be an integer, forcing ϵ to be rational. Hence both determinants are nonzero. $\square$

The phrase “exactly two” counts orientation. The Randers metric is not reversible, so quotienting the two equators by direction would change the dynamical object.

3 Complete boundary atlas

Theorem 3 (rational and singular faces). *Write $\epsilon = p/q$ in lowest terms, where $0 < |p| < q$. Every non-equatorial geodesic has least period $2\pi q$. The two equators instead have least periods*

$$\frac{2\pi q}{q + p}, \quad \frac{2\pi q}{q - p}.$$

At $\epsilon = 0$ the metric is the round Zoll metric and all great circles have least period 2π . Longitude reflection conjugates ϵ to $-\epsilon$ and exchanges the oriented equators. At $|\epsilon| = 1$ global strong convexity fails; $|\epsilon| \geq 1$ is outside the theorem.

Proof. For a non-equatorial frame, equations (6) are necessary and sufficient. Substitution of p/q shows that q divides m , so the least positive return is $2\pi q$. On the equator the shared-axis calculation used in (3) gives the two displayed values; they must not be replaced by the generic period. The zero-wind and reflection claims follow directly from the indicatrix. Finally, at the equator, where $|W|_h$ attains its maximum, the strict navigation inequality fails when $|\epsilon| = 1$. $\square$

An equator on the rational face is degenerate exactly when $q/(q \pm p)$ is an integer. This is a source resonance statement, not an arithmetic-prime correspondence. This section is the *rational boundary owner* added in revision round one.

Source boundary

This is a source-local reconstruction, not a priority claim. Katok owns the integrable examples; Ziller owns their geometric closed-geodesic analysis; Bao–Robles–Shen own the navigation curvature framework. The finite JSON ledger is an implementation receipt created for this package.

References

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